

2020 AMC 10B Problem 23 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 23

Problem:

Square $ABCD$ in the coordinate plane has vertices at the points $A(1, 1)$, $B(-1, 1)$, $C(-1, -1)$, and $D(1, -1)$. Consider the following four transformations:

- L , a rotation of 90° counterclockwise around the origin;
- R , a rotation of 90° clockwise around the origin;
- H , a reflection across the x -axis; and
- V , a reflection across the y -axis.

Each of these transformations maps the square onto itself, but the positions of the labeled vertices will change. For example, applying R and then V would send the vertex A at $(1, 1)$ to $(-1, -1)$ and would send the vertex B at $(-1, 1)$ to itself. How many sequences of 20 transformations chosen from $\{L, R, H, V\}$ will send all of the labeled vertices back to their original positions? (For example, R, R, V, H is one sequence of 4 transformations that will send the vertices back to their original positions.)

Answer Choices:

- A. 2^{37}
- B. $3 \cdot 2^{36}$
- C. 2^{38}
- D. $3 \cdot 2^{37}$
- E. 2^4

Solution:

Let $(+)$ denote counterclockwise/starting orientation and $(-)$ denote clockwise orientation. Let 1, 2, 3, and 4 denote which quadrant A is in. Realize that from any odd quadrant and any orientation, the 4 transformations result in some permutation of $(2+, 2-, 4+, 4-)$.

The same goes that from any even quadrant and any orientation, the 4 transformations result in some permutation of $(1+, 1-, 3+, 3-)$. We start our first 19 moves by doing whatever we want, 4 choices each time. Since 19 is odd, we must end up on an even quadrant. As said above, we know that exactly one of the four transformations will give us $(1+)$, and we must use that transformation.

Thus, the answer is

$$4^{19} = (C) \boxed{2^{38}}$$

OR

Notice that any pair of transformations either swaps the x and y -coordinates, negates the x and y -coordinates, swaps and negates the x and y coordinates, or leaves the original unchanged. Furthermore, notice that for each of these results, if we apply another pair of transformations, one of these results will happen again, and with equal probability. Therefore, no matter what state we are in after we apply the first 9 pairs of transformations, there is a $\frac{1}{4}$ chance the last pair of transformations will return the figure to its original position.

Therefore, the answer is

$$\frac{4^{20}}{4} = 4^{19} = (C) \boxed{2^{38}}$$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) 